

AIGR Deep Research Report — *S. pombe* *sgo1* (Q9P7A0)

Hypothesis (slug: mitotic-evidence-provenance): Determine whether

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establishes an *endogenous mitotic function* for *S. pombe* Sgo1, and distinguish this from engineered expression or experiments on other shugoshins (Sgo2).

Annotation under review (PomBase, retained): `sgo1` → GO:1990758 "mitotic sister chromatid biorientation", evidence IMP, reference

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annotation extension `happens_during` GO:0000089 "mitotic metaphase". (This is the "mitotic-metaphase IMP annotation" named in the task context.)

Executive Judgment

Verdict: Over-annotated for Sgo1 (weakly supported at best). The endogenous mitotic function in

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is established for the paralog Sgo2, named explicitly in Figure 2.

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(Tsukahara, Tanno & Watanabe, *Nature* 2010, 467:719–723, "Phosphorylation of the CPC by Cdk1 promotes chromosome bi-orientation") is a genuine *S. pombe* **mitotic** study. Its endogenous mitotic mechanism — Cdk1–cyclin-B phosphorylation of Survivin (Bir1) driving direct binding to shugoshin, which acts as the centromeric adaptor recruiting the chromosomal passenger complex (CPC) for bi-orientation — is demonstrated for **Sgo2**, the mitotic fission-yeast shugoshin.

Figure-level provenance (recovered in Iteration 2). The publisher HTML (doi:10.1038/nature09390) exposes the abstract, all four figure captions, and the reference list. **Figure 2 is titled: "Phosphorylation of Bir1 promotes the association with Sgo2 and targeting the CPC to centromeres."** Figures 1, 3 and 4 concern Bir1 phosphorylation, human Borealin, and generic "shugoshins." Across **all accessible text, the string "Sgo1" occurs 0 times, while "Sgo2" occurs 4 times** (the Fig. 2 title plus citations to human *Tripin/hSgo2*). Thus the named fission-yeast shugoshin in the paper's figures is **Sgo2, not Sgo1**.

This matches the paper's own PomBase curation (a Canto session curated by author Y. Watanabe), which assigns every piece of **direct** mitotic evidence — adaptor activity (GO:0140463 IPI, "recruits Bir1"), protein–protein binding, and two-hybrid interaction phenotypes — to **Sgo2 (SPAC15A10.15)**. Sgo1 (SPBP35G2.03c) received *only* a bare gene-level GO:1990758 **IMP** with **no allele, no genotype, and no accompanying mutant-phenotype (FYPO) annotation** — unusual for an IMP, which by definition requires a mutant phenotype.

Independently, Sgo1 is **meiosis-specific** (Kitajima 2004; Kawashima 2007; Vanoosthuyse 2007) and its PomBase expression records are entirely meiosis-associated, so an *endogenous* mitotic role is biologically implausible. The most parsimonious reading is that the Sgo1 IMP is a **paralog-parallel / conserved-mechanism carry-over annotation** rather than a figure-supported endogenous mitotic function of Sgo1.

Most important caveat (reported honestly): The *Nature* full body text and **Supplementary remain paywalled** (not in PMC/Europe PMC/OA; Semantic Scholar returned no open PDF). I could read the abstract, figure captions, and reference list but **not the Methods/Supplementary**, so a buried Sgo1 mention cannot be *fully* excluded. However, the main figure explicitly names Sgo2 and Sgo1 is absent from all accessible content, so the residual uncertainty is small. Even if a minor in-vitro Sgo1 binding experiment existed, it would support a **biochemical/conserved-binding** property (an MF/IPI statement), not an **endogenous in-vivo mitotic BP** role.

Evidence Matrix

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	P 20739936 (Tsukahara 2010, <i>Nature</i>) — abstract	Direct assay (in vivo + biochemical)	Qualifies hypothesis	Does the paper show a mitotic shugoshin→CPC adaptor function?	Cdk1-phosphorylated Survivin binds "shugoshin," defined as a conserved centromeric CPC adaptor; enables mitotic bi-orientation	<i>S. pombe</i> mitosis (prometaphase/ metaphase)
1b	P 20739936 — Figure captions (publisher HTML, doi:10.1038/nature09390)	Direct assay (figure provenance)	Refutes Sgo1-specific reading; Supports Sgo2	Which shugoshin is named in the fission-yeast experiment?	Fig. 2 title: "Phosphorylation of Bir1 promotes the association with Sgo2 and targeting the CPC to centromeres." "Sgo1" occurs 0×, "Sgo2" 4× in all accessible text	<i>S. pombe</i> mitosis
2	P 20739936 — PomBase Canto curation of this paper (curator: Y. Watanabe; 9 genes)	Review/ database (structured author curation)	Refutes Sgo1-specific reading; Supports Sgo2	Which shugoshin carries the direct mitotic evidence?	Adaptor activity GO:0140463 (IPI, "recruits Bir1") , reciprocal protein-binding IPI Bir1↔Sgo2 , and two-hybrid "abolished PPI" phenotypes are all on Sgo2 (SPAC15A10.15) . Sgo1 gets only a bare GO:1990758	<i>S. pombe</i> mitosis

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					IMP (no allele, no FYPO phenotype).	
3	P 17322402 (Kawashima 2007, <i>Genes Dev</i>)	Mutant phenotype + interaction	Refutes Sgo1 mitotic role	Which shugoshin is mitotic and binds Bir1/Survivin?	"Sgo1 is meiosis specific"; Sgo2 interacts with Bir1/Survivin and promotes Aurora/CPC localization for tension-generating attachment	<i>S. pombe</i> mitosis & meiosis
4	P 17301288 (Vanoosthuyse 2007, <i>J Cell Sci</i>)	Mutant phenotype/ localization	Refutes Sgo1 mitotic role	Which shugoshin ensures mitotic biorientation?	"Sgo1 ... protector of centromere cohesion in meiosis I"; Sgo2 required for mitotic biorientation and CPC/passenger retention	<i>S. pombe</i> mitosis
5	P 14730319 (Kitajima 2004, <i>Nature</i>)	Mutant phenotype	Refutes / paralog context	Original identification of Sgo1 vs Sgo2 roles	Sgo1 protects meiotic Rec8 cohesin; Sgo2 paralog required for faithful mitotic segregation	<i>S. pombe</i>
6	PomBase gene record SPBP35G2.03c (expression + paralog)	Database	Qualifies/ Refutes	Is Sgo1 expressed in mitosis?	Sgo1 expression annotations are meiosis-associated only ; Sgo1 is the paralog of Sgo2 ;	<i>S. pombe</i>

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					all other Sgo1 BP terms are meiotic	

GO Curation Implications (leads requiring curator verification)

Target annotation: `sgo1` GO:1990758 (mitotic sister chromatid biorientation), IMP,

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`happens_during` GO:0000089.

- **Aspect:** BP (biological process).
- **Recommended action (lead): Flag for review; likely too strong / candidate for removal or re-scoping to the paralog Sgo2.** The evidence code **IMP is not adequately supported for Sgo1** because the curation carries no Sgo1 allele or mutant phenotype, and all direct experimental evidence in the paper is Sgo2-specific. Sgo1 is meiosis-specific.
- **Preferred outcomes, in order:**
 - **Remove** the Sgo1 GO:1990758 IMP if figure review confirms no Sgo1 loss-of-function mitotic phenotype exists (most likely).
 - If the paper contains an **in-vitro** demonstration that recombinant Sgo1 binds phospho-Survivin (conserved-mechanism exemplar), **re-scope** to a molecular-function/interaction statement (e.g., IPI protein binding to Bir1/Survivin) **without** an in-vivo mitotic BP claim, and add a note that Sgo1 is meiosis-specific.
 - If retained, add an explicit provenance comment distinguishing it from the endogenous Sgo2 function.
 - **Do NOT** default to "protein binding" as the informative term for the mitotic role; the informative mitotic MF term (**GO:0140463 chromatin-protein adaptor activity, recruits Bir1**) is correctly held by **Sgo2**, and Sgo1's core, well-supported terms are meiotic (GO:1990813; GO:0045144; GO:0051455) plus centromere/heterochromatin CC.

Mechanistic Scope

- **Immediate molecular event tested by the paper:** Cdk1–cyclin-B phosphorylation of Survivin (Bir1) creates a phospho-dependent binding surface for shugoshin; shugoshin then acts as the centromeric adaptor that docks the CPC (Aurora B/Ark1–INCENP/Pic1–Survivin/Bir1–Borealin/Nbl1) at centromeres to enable bi-orientation.
 - **Direct gene-product activity (Sgo2):** phospho-Survivin binding + CPC recruitment (chromatin-protein adaptor activity) — a *direct* mitotic molecular function.
 - **Downstream/indirect:** chromosome bi-orientation and faithful mitotic segregation are downstream cellular outcomes of correct CPC localization.
 - **Sgo1:** no direct mitotic molecular activity is demonstrated; its established direct activity is meiotic centromeric cohesin (Rec8) protection with PP2A. Any Sgo1 signal in this paper is at most a **conserved biochemical binding property**, not an endogenous mitotic process role.
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Conflicts and Alternatives

- **Paralog confusion (primary alternative):** "Shugoshin" in the abstract is generic; the mitotic shugoshin in *S. pombe* is **Sgo2**. The Sgo1 annotation risks conflating the paralogs. The paper's own curation localizes direct evidence to Sgo2.
 - **Engineered/conserved-mechanism exemplar:** Watanabe's group frequently uses the small (319-aa) Sgo1 coiled-coil as a tractable model of the conserved shugoshin CPC-binding module. An in-vitro Sgo1–phospho-Bir1 binding result would be a **conserved-mechanism** demonstration, not endogenous mitotic biology.
 - **Author-curated over-annotation:** Because the senior author curated the paper, the Sgo1 IMP may reflect an intent to flag conserved mechanism at the gene level; the missing allele/phenotype supports "carry-over/parallel annotation" rather than a data-backed IMP.
 - **No conflicting evidence** was found that Sgo1 is expressed or functions in vegetative mitosis.
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Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Exact figure/ panel that generated the Sgo1 IMP	Largely resolved (Iter. 2-3): Fig. 2 names Sgo2; "Sgo1" absent from all accessible text (abstract, 4 captions, reference list, editorial summary). The paywalled Methods are not in the HTML. The Supplementary PDF is freely downloadable (Suppl. Table 1 + Suppl. Figs 1-18 w/ legends) but its text is Flate-compressed (300 streams) and could not be extracted in the sandbox (no <code>zlib</code>); raw-byte search returned 0 for all terms incl. Survivin/Bir1, so it is inconclusive, not negative. Per the task's own context this is a "Sgo2-focused supplement"	Determines whether any Sgo1-specific panel exists	Decompress/OCR the Supplementary PDF, or read the paywalled Methods, to fully exclude a buried Sgo1 panel (residual, low-probability)
Whether an sgo1 mutant shows a mitotic biorientation phenotype	PomBase FYPO annotations for this paper — none on sgo1	IMP requires a mutant phenotype; its absence undermines the IMP	Locate any sgo1Δ / sgo1 point-mutant mitotic assay in the paper (none evident)
Sgo1 protein presence in mitosis	PomBase expression = meiosis-only	Endogenous mitotic function requires mitotic expression	Vegetative-cell proteomics/Western or tagged-Sgo1 imaging in mitosis

Discriminating Tests

1. Full-text figure audit of

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(and Supplementary): catalog every panel labeled "Sgo1" vs "Sgo2," recording whether it is in-vitro binding, localization, or an in-vivo mutant phenotype. This single step is decisive.

2. **sgo1Δ vs sgo2Δ mitotic bi-orientation assay** (minichromosome loss / lagging chromosomes / CPC-Bir1 centromere localization in vegetative cells): predicts a defect only in sgo2Δ.
3. **Mitotic expression check** for endogenous Sgo1 (tagged strain, vegetative growth): predicts undetectable Sgo1 in mitosis.
4. **In-vitro phospho-Bir1 pulldown** comparing recombinant Sgo1 vs Sgo2 N-terminal coiled-coils: tests whether Sgo1 retains conserved binding (a biochemical property) even though it is not expressed in mitosis.

Curation Leads (require curator verification)

- **Candidate reference to verify with exact snippet:**

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abstract — "*Survivin phosphorylation promotes direct binding with shugoshin, which we now define as a conserved centromeric adaptor of the CPC.*" Verify in the figures **which shugoshin** (Sgo1 vs Sgo2) was used for each assay.

- **Candidate action change:** Re-scope or remove `sgo1` GO:1990758 (IMP,

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the mitotic biorientation/CPC-adaptor function is supported for **sgo2** (GO:0140463 IPI recruits Bir1; GO:1990758 IMP) in the same paper.

- **Candidate corrected placement:** If a real Sgo1 experiment exists, it is most likely an **in-vitro conserved-binding** result → an MF/interaction statement (Sgo1–Bir1/Survivin binding), **not** an in-vivo mitotic BP.
- **Suggested curator questions:**

- Is there any sgo1 allele or loss-of-function mitotic phenotype in

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(required to justify IMP)?

- Was recombinant Sgo1 used only as a conserved-mechanism exemplar?
- **Suggested experiments:** the four discriminating tests above.
- **Retain (well-supported) Sgo1 core terms:** meiotic centromeric cohesion protection (GO:1990813), meiotic sister chromatid segregation (GO:0045144), spindle attachment to meiosis I kinetochore (GO:0051455); CC centromere/pericentric heterochromatin/inner kinetochore.

Limitations

- Figure captions were recovered directly from the publisher HTML (Iteration 2): **Fig. 2 explicitly names Sgo2**, and "Sgo1" is absent from all accessible text. The *Nature Methods/ full body remain paywalled*. The **Supplementary PDF is freely downloadable but text-locked** (Flate-compressed; not extractable in the sandbox, no `zlib`), so it was neither confirmed nor excluded programmatically — the task context itself labels it "Sgo2-focused." A buried Sgo1 mention in Methods/Supplementary therefore cannot be 100% excluded (residual, low probability).
- Conclusions rest on the paper abstract + **figure captions**, the structured PomBase author curation of this exact paper (direct evidence all on Sgo2; Sgo1 = bare IMP, no allele/phenotype), and orthogonal primary literature (Kitajima 2004; Kawashima 2007; Vanoosthuyse 2007). These lines of evidence are mutually consistent.